

This vs That: The Economic Effects of Fiscal Spending Shifts

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Abstract

In this paper, we investigate how redistributing government spending across different spending categories influences the economy. An exogenous increase in spending on a component results in a change in total spending and also a shift in the distribution of spending. Our VAR model decomposes this composite shock into a level shock and a reallocation shock. This allows us to estimate the macroeconomic implications of redistributing spending between two components without a corresponding increase in total spending. We find that a reallocation towards government investment from government consumption raises output in the short run. Conversely, a shift toward defense spending from non-defense spending does not produce a significant change in output. Additionally, a higher share of state and local government purchases is beneficial in the short-term.

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1 Introduction

Recently, there has been a revival of interest in the assessment of fiscal policy measures. This renewed interest stems from the realization, by the central banks of advanced economies, of the limitations of monetary policy (zero lower bound) in reviving their respective economies since the Global Financial Crisis of 2008. For example, since the Covid-19 crisis, the U.S. government passed three main relief packages and one supplemental package, totaling nearly \$2.3 trillion (around 11% of its GDP).¹ Due to the increased reliance on fiscal policy to boost economic growth during a downturn, there has been a surge in research that quantifies the effects of government spending.

Primarily, past studies have considered total government expenditure as a homogeneous good (Li, 2014). For example, Blanchard and Perotti (2002), Mountford and Uhlig (2009) and Ramey (2011), each apply different methodologies to identify government spending shocks while adhering to the aforementioned assumption. However, government spending spans different critical areas like military, infrastructure, healthcare, etc. Recent literature distinguishes between the fiscal multipliers arising from different categories of spending. The aggregate multiplier for the total government purchases would be approximately a weighted average of the multipliers of the components (Boehm, 2020). The effectiveness of the fiscal policy therefore does not only depend on the size but also on the composition of the basket of products purchased by the government (Ramey and Shapiro, 1998; Muratori et al., 2023). Over the years, there has been a shift in the composition of the purchases made by the U.S. government.² In this paper, we investigate how a cross-sectional shift in the spending distribution affects economic outcomes.

When an exogenous shock affects spending on any component, it also influences spending on other components due to underlying correlations, thus altering the total spending level. This change, if substantial, leads to a shift in the proportion of spending across components, modifying their weights within the government’s expenditure basket. For instance, a substantial rise in military expenditures driven by escalating international conflicts, such as those in the Middle East, the war in Ukraine, and other global security threats, might cause a decrease in the proportion of spending allocated to education and healthcare. The GDP response to such a shock reflects both the increase

¹<https://www.imf.org/en/Topics/imf-and-covid19/Policy-Responses-to-COVID-19>

²Federal defense spending declined from 50% of total government purchases in 1960, with defense consumption at 35% and investment at 15%, to 23% in 2020—18% for consumption and 5% for investment, as detailed in Figures 1.1 and 1.2. Conversely, state and local government spending has increased, primarily driven by consumption expenditures, which rose from 28% of the total in 1960 to nearly 50% in 2020, while their investment has consistently ranged between 10-12%.

in aggregate spending and the change in its allocation. In prevailing VAR specifications that include individual spending components, these two channels cannot be separated.

We take a different approach that directly estimates the output consequences of reallocating spending from one component to another, a question that maps naturally to the budget allocation decisions that policymakers routinely face. To do so, we specify a structural VAR that includes total government spending and the share of one component in the total. We impose the restriction that the share does not respond contemporaneously to a shock to total spending, and that total spending does not respond contemporaneously to a shock to the share. A shock to total spending therefore represents a pure level effect in which all components increase proportionally. A shock to the share represents a pure reallocation in which spending shifts from one component to another while aggregate spending remains unchanged on impact.

In our analysis, we evaluate the effects of shifting spending across three pairs of components, examining each pair independently: (i) government investment and consumption, (ii) defense and non-defense spending, and (iii) state and local government spending versus federal government spending. The level shock to total spending in each specification yields multipliers in the range of 0.6 to 1.2 over a five-year horizon, consistent with the existing literature.³ Despite each specification applying a shock to the same total government spending, the implied multipliers differ across specifications. This suggests that the aggregate spending multiplier is not independent of the underlying allocation of government purchases.

Analyzing the reallocation shocks within each pair, we find that a shift toward government investment raises public investment and reduces government consumption, while total spending increases modestly over time. GDP rises on impact, producing a multiplier of approximately 2. Although the output effect is concentrated in the short run, it underscores the beneficial impact of investment in public capital.

Decomposing total government purchases into military and non-military expenditures, we find that a reallocation toward defense spending raises military expenditures but lowers non-defense spending. The output response is negative but not significantly different from zero, with the multiplier over a five-year horizon equal to -0.93. This indicates that a reallocation toward defense

³Mountford and Uhlig (2009) reports a government spending multiplier of 0.65 over a 1-quarter horizon and -2.07 over 20 quarters, while Auerbach and Gorodnichenko (2012) finds a 0.57 multiplier over 20 quarters. Ramey (2011) estimates multipliers ranging from 0.6 to 1.2, depending on the sample, and Blanchard and Perotti (2002) reports peak multipliers between 0.9 and 1.29.

spending does not produce a significant change in output.

Lastly, a reallocation toward state and local government spending from federal spending raises output in the short run, with the multiplier reaching 3.51 on impact and remaining statistically significant throughout the horizon. The five-year multiplier is 2.06. This likely reflects the large share of public investment in state and local budgets and their ability to deploy federal grants quickly through existing procurement channels.

The paper provides empirical evidence that emphasizes the critical role the allocation of government purchases plays in enhancing fiscal policy efficacy. Our findings show that a cross-sectional shift in spending can lead to significant macroeconomic consequences. A shift from a less productive to a more productive component results in positive economic outcomes, enabling this framework to determine which component is more efficient.

The rest of the paper is organized as follows: Section 2 reviews the relevant literature that grounds our research. Section 3 details our empirical setup and section 4 covers the data utilized. Section 5 presents and discusses the results, and section 6 concludes.

2 Related Literature

This paper is related to the extensive literature that concentrates on the inherent features of fiscal spending for quantifying the effectiveness of the fiscal policy. Several studies (Perotti, 2004; Pappa, 2009; Auerbach and Gorodnichenko, 2012; Ilzetzi et al., 2013; Ellahie and Ricco, 2017; Klein and Linnemann, 2023) compare the macroeconomic outcomes from two or more spending components. Different components result in heterogeneous macroeconomic implications underscoring the importance of the composition of government spending. This paper contributes to the literature by providing an empirical framework that computes dynamic responses from shifts in the composition of spending. Utilizing national time series data, our findings highlight that changes in the proportion of spending in different categories can result in changes in economic output.

The paper is also related to several strands of literature that evaluate the components within each of the three pairs considered in our analysis: (i) government investment and consumption, (ii) defense and non-defense spending, and (iii) state and local government spending versus federal government spending. Baxter and King (1993) incorporate government investment as an unpaid factor in the production function by viewing public capital as an input that directly enhances the

productivity of private capital and labor. However, the short-run output multiplier may respond less to government investment than to government consumption due to the negative wealth effect of increased taxation and implementation delays (Leeper et al., 2010; Ramey, 2021). Additionally, Boehm (2020) finds that short-lived investment shocks crowd out private investment more than government consumption. This occurs because the long service life of private capital leads to a very high intertemporal elasticity of substitution in investment demand. In contrast, with long-lived investment shocks, households have less flexibility to delay their investment decisions. Furthermore, if public capital is productive, then the long-run multipliers can be sizeable (Ramey, 2021). Moreover, Auerbach and Gorodnichenko (2012), Ellahie and Ricco (2017) and Laumer and Philipps (2020) find higher multipliers for government investment compared to government consumption. Our approach focuses on the implications of shifting the share of spending within categories, rather than merely increasing the spending level of a category, as the latter does not necessarily imply a change in its proportion. An increase in the government investment share results in a short-lived increase in output, indicating that government investment is more productive than government consumption in the near term.

Several studies investigate the multipliers from military and non-military expenditures. Military expenditures do not contribute significantly to the production or utility functions of an economy and primarily impact through a negative wealth effect, leading to reduced private sector GDP (Baxter and King (1993)). This type of spending decreases private consumption and investment, resulting in a multiplier that is often less than one and typically lower than those for non-defense spending (Auerbach and Gorodnichenko, 2012; Barro and De Rugy, 2013; Ellahie and Ricco, 2017; Laumer and Philipps, 2020). Our findings indicate that increasing the share of defense spending reallocates funds from non-defense spending without producing a significant change in output. The reallocation multiplier is negative but statistically insignificant, suggesting that at this level of aggregation, defense and non-defense spending do not differ materially in their effects on output.

Lastly, state and local government expenditures are pro-cyclical due to balanced budget requirements, resulting in multipliers below 1, consistent with Ricardian effects where households anticipate future taxes to repay government debt, leading them to save rather than spend (Clemens and Miran, 2012). Unlike federal government spending, state and local governments allocate a smaller proportion to defense, leading to higher multipliers for their expenditures (Ellahie and Ricco, 2017; Laumer and Philipps, 2020). If the share of state and local government spending is to increase

through larger federal grants without an accompanying change in total government expenditures, then output increases in the short run. The multiplier is positive and statistically significant throughout the horizon, suggesting that state and local spending is more productive than federal spending at generating output. Our approach helps compare spending categories by analyzing output changes from redistribution, offering new insights for policy-making on the economic effects of changes in composition.

3 Econometric Methodology

3.1 Prevailing Models

Let $G_{i,t}$ represent the fiscal spending in category i at time t . Assuming $i = 1, 2$ and $G_t = G_{1,t} + G_{2,t}$ denote the aggregate government spending. Prevailing empirical studies (Blanchard and Perotti, 2002; Perotti, 2004; Auerbach and Gorodnichenko, 2012; Ilzetzki et al., 2013; Klein and Linnemann, 2023) that assess and compare the macroeconomic effects stemming from two different components employ a basic VAR specification in the following form:

$$\begin{bmatrix} G_{1,t} \\ G_{2,t} \\ Z_t \end{bmatrix} = C + \Phi(L) \begin{bmatrix} G_{1,t-1} \\ G_{2,t-1} \\ Z_{t-1} \end{bmatrix} + \begin{bmatrix} v_t^{G_1} \\ v_t^{G_2} \\ v_t^z \end{bmatrix} \quad (1)$$

$$v_t \sim N(0, Q)$$

where C denotes a constant term and Z_t is an $M \times 1$ vector of macroeconomic variables like government revenue, GDP, employment, etc. $\Phi(L)$ represents the lag operator polynomial and v_t is the vector of reduced form residuals with non-zero correlations. The objective is to assess how an exogenous change in spending on component G_i influences the aggregates in Z .

A shock to G_i also affects the spending on the other component $G_{i \neq j}$ and changes the aggregate spending (G). The resulting GDP response therefore reflects both a volume effect from higher total spending and a composition effect from the altered distribution across components. These two channels cannot be separated within this specification.⁴

⁴Ellahie and Ricco (2017) incorporates six components in their SVAR specification. The number of components does not resolve this problem. Disentangling volume from composition effects remains difficult with the standard VAR specification.

To see why, consider a positive shock to G_i . Total government spending G will increase, but spending on $G_{i \neq j}$ may rise, fall or remain unchanged. If $G_{i \neq j}$ also rises, the share of G_i in total spending may actually decline despite the positive shock to its level. Conversely, if $G_{i \neq j}$ falls, the share of G_i rises, but the net change in total spending is ambiguous. In either case, inferring the change in spending composition from the impulse responses of individual components is not straightforward. An unexpected increase in spending on G_i does not necessarily imply a reallocation of spending toward that category.

We therefore take a different approach. Rather than estimating the effects of shocks to individual spending components, we aim to estimate the response of Z to shifts in the composition of government spending across categories.

3.2 Proposed Model Specification - Two Components

Considering the previously defined components of government spending, $G_{1,t}$ and $G_{2,t}$, with $G_t = G_{1,t} + G_{2,t}$, we define $s_{1,t} = G_{1,t}/G_t$ as the share of the first component $G_{1,t}$ in the total spending G_t . To capture shifts in the composition of government spending, we rewrite the VAR as follows:

$$\begin{bmatrix} G_t \\ s_{1,t} \\ Z_t \end{bmatrix} = C + \Phi(L) \begin{bmatrix} G_{t-1} \\ s_{1,t-1} \\ Z_{t-1} \end{bmatrix} + \begin{bmatrix} v_t^G \\ v_t^{s_1} \\ v_t^z \end{bmatrix} \quad (2)$$

$$v_t \sim N(0, Q)$$

As in the model discussed previously, C represents a constant term and Z_t is the $M \times 1$ vector that includes macroeconomic variables such as government revenue, GDP, etc. $\Phi(L)$ represents the conformable lag polynomial of finite order p and v_t is a $(2 + M) \times 1$ vector of reduced form residuals that are normally distributed with a mean 0 and a non-diagonal variance covariance matrix Q .

A shock to s_1 implies an increase in s_1 and an equal decrease in $s_2 = 1 - s_1$. However, this does not by itself constitute a reallocation of spending from G_2 to G_1 . If total spending also changes in response to the share shock, the decline in s_2 could simply reflect G_2 growing more slowly than total spending G , rather than an actual reduction in G_2 . To isolate a pure reallocation, we restrict total spending from responding contemporaneously to the share shock. Under this restriction, a rise in s_1 implies that spending shifts from G_2 to G_1 while aggregate spending remains unchanged

on impact. By the same logic, a shock to G with the share held fixed contemporaneously provides a pure level effect in which all components increase proportionally. This specification allows us to decompose the composite shock to an individual component G_1 in equation (1) into two distinct channels: a level shock that changes the total volume of spending and a reallocation shock that changes its composition.

3.3 Bayesian Estimation

We adopt a Bayesian approach to estimate the VAR model specified in equation (2), treating the coefficients Φ and the variance covariance matrix Q as random variables. This allows us to incorporate prior information and to capture parameter uncertainty when constructing impulse responses and multipliers. Full details of the Bayesian estimation procedure, including the prior specification, posterior updating, and hyperparameter choices, are relegated to Appendix B.

Estimation proceeds via a Gibbs sampler that iteratively draws from the conditional posterior distributions of the model parameters. In each iteration, we first draw the VAR coefficients Φ from their conditional posterior distribution, given the current draw of Q and the data. Draws that imply non-stationary dynamics are discarded to ensure stability of the VAR in each regime (Bernanke et al., 2005). We then draw the variance covariance matrix Q from its conditional posterior distribution, given the new draw of Φ . Rather than drawing Q directly, we decompose it into a lower triangular matrix L and a diagonal matrix D , and draw their elements separately. This decomposition allows us to impose the identifying restrictions necessary to distinguish between level and reallocation shocks, as described in the following subsection. We run the sampler for 30,000 iterations and discard the first 15,000 as burn-in.

3.4 Impulse Responses to Fiscal Shocks

To recover structural shocks from the reduced-form residuals, we decompose the $N \times N$ variance covariance matrix Q following the approach in Chan and Jeliazkov (2009):

$$\begin{aligned} Q &= L^{-1} D L^{-1'} \\ v_t &= L^{-1} D^{1/2} \varepsilon_t \\ \varepsilon_t &\sim N(0, I) \end{aligned} \tag{3}$$

where ε_t is an $N \times 1$ vector of mutually independent structural shocks, D is a diagonal matrix with positive diagonal elements, and L is a lower triangular matrix with ones on the main diagonal. The structural shocks are related to the reduced-form residuals through the matrix L^{-1} , whose off-diagonal elements govern the contemporaneous transmission of shocks across variables.

We impose the restriction that the element $L(2, 1)$ equals zero. This ensures that a shock to total spending G does not contemporaneously alter the composition of spending, so that the level shock represents a proportional increase across all components. Furthermore, since G is ordered before s_1 , total spending does not respond contemporaneously to a shock to the share. A surprise increase in s_1 therefore represents a pure reallocation of spending from one component to the other at constant aggregate spending. Together, these two restrictions allow us to cleanly separate volume effects from composition effects. Also, the restriction $L(2, 1) = 0$ renders the ordering between G and s_1 immaterial. Reversing their order while maintaining the zero restriction yields an identical identification, as the two variables are contemporaneously independent in either case.

3.5 Government Spending Multiplier

The multiplier is commonly used in the literature to assess the impact of a dollar increase in fiscal spending on overall output. It is useful for analyzing different fiscal policy scenarios and aiding in policy making. One approach is to measure the ratio of the output response to the initial change in the fiscal spending component, known as the impact multiplier (Mountford and Uhlig, 2009). Blanchard and Perotti (2002) define the peak multiplier as the maximum output response parameter divided by the initial change in government spending. However, this method focuses on the change in output at a particular point in time and may not capture subsequent negative output responses. Ramey (2011) calculates the fiscal multiplier using the integral under the impulse response function. To enable a comparison with the findings of Auerbach and Gorodnichenko (2012), who report multipliers for disaggregated fiscal components, we employ their approach. We then calculate the multiplier in response to the level (G) over the entire path of the response upto a period K , referred to as the sum multiplier, as follows:

$$Level\ Multiplier(K) = \frac{\sum_{\kappa=0}^K y_{\kappa}}{\sum_{\kappa=0}^K G_{\kappa}} \times \frac{\bar{y}}{\bar{G}} \quad (4)$$

Here, y_{κ} and G_{κ} are the output and the government spending responses for a particular period

κ . $\frac{\bar{y}}{G}$ is the sample mean of the GDP to government spending ratio. Since we consider these variables in natural logarithms, the impulse responses represent elasticities and are thus scaled by the sample mean ratio. For a shock to the total level of spending (G), we apply the above formula.

When calculating the sum multiplier resulting from a reallocation shock, i.e., a shock to G_1 's share (s_1), we obtain the impulse response of G_1 by summing the impulse responses of the total level G and share of the component s_1 . This is because both are included in the VAR specification in natural logarithms. Using $G_{1,\kappa}$ to denote the response of component G_1 in period κ to the reallocation shock, the multiplier can be computed as follows:

$$\text{Reallocation Multiplier}(K) = \frac{\sum_{\kappa=0}^K y_{\kappa} \times \frac{\bar{y}}{s_1}}{\sum_{\kappa=0}^K G_{1,\kappa} \times \frac{\bar{G}_1}{s_1}} \quad (5)$$

In this equation, $\frac{\bar{y}}{s_1}$ represents the sample mean of the GDP to the share of the fiscal component, G_1 and $\frac{\bar{G}_1}{s_1}$ is the average ratio of spending in G_1 to its share, s_1 . The numerator quantifies the total output response in dollars per additional unit change in the share across K quarters. Likewise, the denominator calculates the change in spending for component G_1 in dollars per additional unit change in its share over the same period. This multiplier then reflects the dollar change in output over period K for each dollar increase in spending on the component.

We compute the level and reallocation multipliers for every $(d+1)^{th}$ iteration using the impulse responses corresponding to that iteration. We report the median multiplier for different values of K along with the respective 68% credible sets in all cases.⁵

4 Data

The analysis covers the period from the first quarter of 1960 to the fourth quarter of 2022. We use quarterly data on government purchases from the National Income and Product Accounts (NIPA), published by the Bureau of Economic Analysis. All series are converted into real per capita terms using the GDP deflator and population. Total government spending (G) is defined as the sum of

⁵ Additionally, Mountford and Uhlig (2009) suggests taking into consideration the discounted values of the impulse responses to compute the sum multiplier as follows: *Present Value Multiplier* (K) = $\frac{\sum_{\kappa=0}^K (1+\bar{r})^{-\kappa} y_{\kappa}}{\sum_{\kappa=0}^K (1+\bar{r})^{-\kappa} G_{\kappa}} \times \frac{\bar{y}}{G}$ where $\bar{r}$ denotes the average federal funds rate over the sample. We use the above formula to compute the discounted sum multiplier in response to the level shock. In order to compute the multiplier from the reallocation shock, we simply adjust equation (12) to include the discounted impulse responses of y and G_1 in response to the reallocation shock. We report these multipliers in Appendix C and find negligible differences.

six underlying components:

1. Federal defense consumption expenditures: compensation of military personnel and purchases of goods and services for current military operations.
2. Federal defense gross investment: spending on military structures, equipment, and defense-related research and development.
3. Federal non-defense consumption expenditures: compensation of civilian federal employees and purchases for the operation of federal agencies and programs.
4. Federal non-defense gross investment: spending on federal civilian structures, equipment, and non-defense research and development.
5. State and local government consumption expenditures: akin to federal non-defense consumption, covers the provision of services at the state and local level
6. State and local government gross investment: similar to federal non-defense investment, centers on localized infrastructure tailored to state and community needs

From these components, we construct three pairs for analysis. The first pair decomposes total spending into government investment, which sums all three investment components, and government consumption, which sums all three consumption components. The second pair decomposes total spending into defense spending, which combines federal defense consumption and investment, and non-defense spending, which comprises all remaining components. The third pair decomposes total spending into state and local government spending and federal government spending.

For each pair, the baseline VAR includes the natural logarithm of real total government purchases per capita and the natural logarithm of the share of one component in the total. The vector Z_t includes two additional variables: the fiscal deficit, defined as the difference between federal government expenditures and receipts as a percentage of nominal GDP, and the natural logarithm of real GDP per capita. Following Blanchard and Perotti (2002), we detrend all variables by removing a quadratic time trend. Further details on variable definitions and data sources are provided in Appendix A.

5 Results

We now turn to estimating the effects of shifts in the composition of government spending on the economy. Specifically, we analyze how an unexpected reallocation of spending from one component to another, holding total spending fixed, affects macroeconomic aggregates. The policy experiment is defined as: given a fixed fiscal envelope, reallocate one dollar from one category of spending to another. This maps directly to a budget reallocation decision that policymakers routinely face. We consider three distinct pairs of spending components and for each pair, we first estimate the impulse responses and multipliers for aggregate spending (level) shocks using the model proposed in Section 3.2. We then estimate the responses to a reallocation shock that shifts spending toward the first component at the expense of the second, while keeping total spending unchanged on impact.

5.1 Government Investment and Consumption

We define G_1 as government investment (GI), which includes federal defense and non-defense investment, as well as state and local government investment. Similarly, G_2 represents government consumption (GC), encompassing federal defense and non-defense consumption, as well as state and local government consumption expenditures. The sum of GI and GC equals the total government spending G . We denote s_1 as the share of government investment and s_2 as the share of government consumption in total government spending (G).

5.1.1 Responses to Total Government Spending (Level) Shock

Figure 2.1 presents the median impulse responses of total government spending (G), the share of government investment (s_1), fiscal deficit, and GDP to a positive shock to total government spending, along with 68% credible bands. Since G and s_1 enter the VAR in natural logarithms, we recover the responses of G_1 by summing their responses. For G_2 , we use log-linearization.⁶ The responses of the two components are displayed in green in the bottom row of Figure 2.1.

Total spending rises by approximately 0.9% on impact and remains close to 0.8% for the first four quarters before declining gradually, reverting to zero by the end of the horizon. The response remains significantly different from zero throughout. The share of government investment (s_1) does not respond contemporaneously, consistent with the identification restriction. It begins to fall after

⁶Since $G_2 = (1 - s_1)G$, we have $\Delta \log G_2 = \Delta \log G + \Delta \log (1 - s_1)$. Applying a first-order approximation around the sample mean of the share, $\bar{s}_1$, $\Delta \log (1 - s_1) \approx -\frac{\bar{s}_1}{1 - \bar{s}_1} \Delta \log s_1$, which gives $\Delta \log G_2 \approx \Delta \log G - \frac{\bar{s}_1}{1 - \bar{s}_1} \Delta \log s_1$

four quarters, with the decline becoming statistically significant around the sixth quarter. The share falls by approximately 0.6% at longer horizons and does not revert to its pre-shock level even as total spending returns to zero. This persistent decline in s_1 suggests that increases in aggregate spending are disproportionately absorbed by government consumption rather than investment.

The responses of the individual components reflect these dynamics. Government investment (GI) increases on impact by 0.7%, driven entirely by the rise in total spending while its share remains unchanged. However, as the share of investment declines persistently, spending on investment also falls, turning negative eventually. Government consumption (GC), by contrast, benefits from both the increase in total spending and the rising share of consumption. Its response closely tracks that of total spending, remaining positive and significant throughout. The fiscal deficit worsens by approximately 0.5 percentage points on impact and remains elevated as long as spending is above its baseline level, improving gradually as total spending reverts to zero.

GDP begins to rise after the shock, reaching approximately 0.1% after the first quarter and maintaining that level through the sixth quarter. The response is statistically significant over this period before fading to zero. The output multipliers from the level shock are positive across all time horizons, as seen in column (1) of Table 1. The multiplier is 0.23 on impact, rises to a peak of 0.71 after eight quarters, and remains statistically significant for horizons 3 to 12. Over a 20-quarter horizon, the average multiplier is 0.58. These estimates are consistent with the existing literature: Auerbach and Gorodnichenko (2012) report a multiplier of 0.57 over a comparable horizon, while Ramey (2011) find estimates ranging between 0.6 and 1.2 depending on the sample.

5.1.2 Responses to Government Investment Share (Reallocation) Shock

We now turn to the macroeconomic effects of a shift in the composition of spending toward government investment. Figure 2.2 presents the impulse responses to a reallocation shock.

The share of government investment rises by approximately 1.7% on impact. This increase partially reverses over the following three quarters, settling around 1% before declining gradually to approximately 0.4% at longer horizons. The response remains statistically significant throughout. Total spending does not respond contemporaneously, consistent with the ordering restriction. It begins to rise after two quarters, increasing steadily to approximately 0.4% at longer horizons. This delayed increase in total spending suggests that the reallocation toward investment generates a modest endogenous expansion in aggregate spending over time.

The responses of the individual components follow directly from these dynamics. Government investment (GI) rises sharply by approximately 1.8% on impact, reflecting the combined effect of the large increase in its share at unchanged total spending. It then declines as s_1 moderates, but remains positive and significant throughout, stabilizing around 1% as total spending begins to rise. Government consumption (GC) falls by approximately 0.5% on impact, as the increase in s_1 implies a corresponding decrease in s_2 . The decline gradually diminishes as total spending rises and the investment share moderates. The response crosses zero around the twelfth quarter and turns slightly positive thereafter, reaching approximately 0.3% at longer horizons where it remains statistically significant.

The fiscal deficit improves by approximately 0.3 percentage points on impact. This initial improvement is consistent with the increase in GDP, which generates higher tax revenues while total spending remains unchanged on impact. However, as total spending rises over subsequent quarters, this improvement dissipates. The deficit reverts to its pre-shock level within a few quarters and turns slightly positive at longer horizons, in line with the persistent increase in aggregate spending.

GDP rises by approximately 0.1% on impact but the increase is short lived, reverting to zero after the first quarter. The corresponding reallocation multiplier is approximately 2.08 on impact, remaining statistically significant for the first quarter before declining. The average multiplier over a 20-quarter horizon is 1.01, though statistically insignificant. Despite the transient nature of the output response, the positive short-run effect suggests that reallocating spending from government consumption to government investment is productive at the margin. This finding is broadly consistent with the literature, which documents larger multipliers for government investment than for government consumption. Auerbach and Gorodnichenko (2012), for instance, report a multiplier of 2.39 for government investment over a comparable horizon. The positive output effect likely reflects the higher immediate demand generated by investment procurement and construction activity, which more than offsets the demand created through the government wage and operational spending channel associated with consumption.

5.2 Defense and Non-Defense Spending

We next examine how shifts in the composition of government spending between military and non-military purchases affect the economy. In this specification, G_1 represents defense spending (DEFS), which comprises federal defense consumption expenditures and federal defense gross investment.

G_2 encompasses all non-defense spending, including federal non-defense purchases and state and local government spending. Together, $G_1 + G_2 = G$ represents total government purchases, with s_1 denoting the share of defense spending in the total. Over our sample period, defense spending accounts for approximately 30% of total government purchases. Given the renewed policy interest in defense budgets amid shifting geopolitical conditions, understanding the macroeconomic consequences of reallocating spending between military and civilian purposes is particularly relevant.

5.2.1 Responses to Total Government Spending (Level) Shock

Total spending rises by approximately 0.9% on impact and declines gradually to about 0.3% at longer horizons, remaining statistically significant throughout. The share of defense spending does not respond contemporaneously, consistent with the identification restriction. It begins to rise in the following quarters, reaching a peak of approximately 0.4% before declining slowly. The increase remains statistically significant for the first 16 quarters. This suggests that aggregate spending shocks disproportionately favor defense expenditures in the short to medium run.

The component responses reflect the combined dynamics of total spending and the defense share. Defense spending (*DEFS*) rises by approximately 0.9% on impact and continues to increase as the defense share rises, peaking at around 1.3% before declining gradually. It remains positive and significant throughout, settling at approximately 0.5% at longer horizons. Non-defense spending (*NDEFS*) rises by approximately 0.8% on impact and traces a path similar to that of total spending, declining gradually to about 0.3% at longer horizons. This is expected, as non-defense spending constitutes approximately 70% of total government purchases over the sample period. Moreover, the shift in composition is modest: a 0.4% increase applied to a defense share of 30% amounts to a change of roughly 0.12 percentage points. With such a small compositional shift, non-defense spending naturally tracks the aggregate response.

The fiscal deficit worsens by approximately 0.5 percentage points on impact. It improves gradually as spending declines but remains elevated at about 0.1 percentage points at longer horizons, consistent with the persistent increase in total spending. GDP rises by approximately 0.1% on impact. Since the composition is held fixed contemporaneously, this initial response reflects a pure volume effect. The output response continues to build, reaching a peak of approximately 0.2% by the fourth quarter before declining. The increase remains statistically significant for 12 quarters.

The cumulative multipliers in response to the level shock, reported in column (1) of Table 2,

are positive and statistically significant across all horizons. The multiplier peaks at 1.14 after eight quarters before declining, and the average multiplier over a five-year horizon is 1.0 and statistically significant.

5.2.2 Responses to Defense Spending Share (Reallocation) Shock

Figure 3.2 displays the responses to a reallocation shock that raises the defense spending share and lowers the non-defense spending share. Initially, s_1 increases by 1.4%. It rises further over the next few quarters and then eases, settling at around 1% by the end of the horizon. Total government spending, G , also increases in response to the shock. Total government spending, G , also rises, peaking at about 0.2% and becoming insignificant only after 16 quarters. As both s_1 and G increase, defense spending rises by 1.4% initially and then increases further, remaining above 1% throughout the horizon. By contrast, because the rise in G is limited, the decline in the non-defense share leads to a fall in non-defense spending of about 0.7% initially. Although the magnitude of this decline narrows over time, non-defense spending remains about 0.4% below baseline.

The fiscal deficit improves on impact following the reallocation shock. However, this improvement disappears within one quarter. The deficit then settles at a negligible increase of roughly 0.1 percentage point for the remainder of the horizon. The GDP response is not significantly different from zero at any horizon. It increases by approximately 0.1% on impact and then turns negative from the first quarter onward, settling near -0.1% for the remainder of the horizon. The implied multiplier is 0.93 on impact and declines to approximately -0.93 over a five-year horizon. These estimates are also not significant. This finding aligns with Auerbach and Gorodnichenko (2012), who report a similarly insignificant defense spending multiplier of -0.21. Although defense spending is primarily directed toward national security and produces goods and services that do not directly enter private production or enhance household consumption, the insignificant output response suggests that its non-defense counterpart is not substantially more productive. A reallocation between these two categories therefore does not produce a measurable change in economic output.

5.3 State and Local Government Spending and Federal Spending

State and local governments are constrained in their ability to independently increase spending, as their expenditures depend largely on own-source revenues. However, they also receive federal grant-in-aid, which tends to be especially large during downturns. In the analysis below, we consider the

effects of reallocating spending from the federal government to state and local governments. In practice, such a reallocation could be achieved through an increase in federal grant-in-aid, directing resources toward the localized consumption and investment needs that state and local governments typically serve.

We define G_1 as state and local government spending (*SLGS*), including state and local government consumption expenditures and gross investment. We define G_2 as total federal government spending (*FGS*), which includes both federal defense and non-defense components. Together, G_1 and G_2 sum to total government purchases, G . Accordingly, s_1 denotes the share of state and local government spending in G , and s_2 denotes the share of federal government spending in G .

5.3.1 Responses to Total Government Spending (Level) Shock

Figure 4.1 displays the responses to a level shock to total government spending. Total government purchases, G , increase by about 0.9% initially, remain positive throughout the horizon, and then ease steadily over time. The share of state and local government spending, s_1 , falls in the early quarters, reaching about -0.2%, but becomes statistically indistinguishable from zero after 8 quarters. This suggests that the relative shares of federal and state and local spending do not change persistently. State and local government spending (*SLGS*) rises by 0.9% initially and then falls to around 0.3%, while remaining significant throughout the horizon. Federal spending also increases by 0.9% at first, peaks at 1.1% by quarter 3 as s_1 declines, and then falls to about 0.3% toward the end of the horizon. As s_1 returns to its initial level, both components move broadly in line with total government spending.

The fiscal deficit rises by about 0.2 percentage point initially and later settles near 0.1 percentage point. GDP increases by 0.3% at first and then returns toward its pre-shock level, with the response becoming statistically indistinguishable from zero after 12 quarters. In line with this response path, the sum multiplier peaks at 1.79 initially and then declines over time, but remains statistically significant at all horizons. The five-year multiplier is 1.18, which lies within the range of 0.6 to 1.2 reported by Ramey (2011).

Our level multipliers across the three specifications are consistent with estimates in the existing literature. However, despite each specification applying a shock to the same total government spending, the GDP responses and implied multipliers differ. The key distinction across specifications is the composition of spending captured by the share variable. Each decomposition pairs total

spending with a different component share, and the resulting dynamics of the sub-components differ in their effects on output. This suggests that the aggregate spending multiplier is not independent of the underlying allocation of government purchases. Changes in the mix of spending categories can influence the size and persistence of the multiplier, even for a given increase in total spending. Differences in multiplier estimates across studies may therefore be partly attributable to differences in the mix of government purchases in their respective samples.

5.3.2 Responses to State and Local Government Spending Share (Reallocation) Shock

We now consider the economic effects of a reallocation shock that shifts government spending from the federal component toward the state and local component.

Figure 4.2 shows that s_1 rises by 0.9% initially in response to its own shock and then declines over time, remaining close to 0.6% in later quarters. Total government spending does not respond significantly to this shock. As a result, state and local government spending (*SLGS*) closely follows the path of s_1 . By contrast, federal government spending (*FGS*) moves in the opposite direction. It falls by 1.2% initially and then gradually recovers, though it remains about 0.6% below baseline in later quarters. In practice, such a shift could occur through larger federal grants-in-aid to state and local governments.

The fiscal deficit declines by 0.4 percentage point initially and then returns to its baseline level by the final quarter considered. The response remains significantly below baseline for about 8 quarters. Since total government spending does not change in response to this reallocation shock, the decline in the fiscal deficit likely reflects the improvement in GDP. GDP increases by 0.3% initially and then moves back toward baseline, remaining significantly above it for about 12 quarters.

The multipliers are positive at all horizons and decline over time, while remaining statistically significant throughout, as shown in column (2) of Table 3. The cumulative multiplier reaches its peak of 3.51 initially and then falls over time. The five-year multiplier is 2.06, which is broadly in line with Ellahie and Ricco (2017), who report a five-year cumulative multiplier of about 2.7 for state and local government consumption and 3.5 for state and local government investment. This is also consistent with Laumer and Philipps (2020), who find larger multipliers for state and local government spending than for federal spending. State and local governments account for a disproportionately large share of public investment, particularly in infrastructure and construction, which tends to generate stronger demand effects due to higher domestic content. Additionally, state and

local governments typically have existing procurement relationships and ongoing projects. Federal grants may therefore translate into actual expenditure more quickly than new federal programs, producing a stronger short-run output response.

6 Conclusion

Since the 2008 financial crisis, fiscal policy has become a pivotal area of focus. A growing literature suggests that different components of government spending produce varied output multipliers, highlighting that the composition of government purchases significantly influences the effectiveness of fiscal measures. Even though the makeup of government expenditures has changed over time, the consequences of these changes remain largely unexplored. This paper investigates the potential effects of variations in the cross-sectional distribution of government spending components, aiming to enhance our understanding of their influence on fiscal policy outcomes.

In this paper, we introduce an empirical approach to analyze the effects of reallocating spending from one component to another. This is accomplished by incorporating the share of one spending component in the total into a structural VAR, along with the sum of the two components. We impose the restriction that the share does not respond contemporaneously to a shock to total spending, and that total spending does not respond contemporaneously to a shock to the share. This separates volume effects from composition effects on impact, allowing us to estimate the output consequences of redistributing spending across components.

We decompose government expenditures into three pairs for analysis: (i) government investment and consumption, (ii) defense and non-defense spending, and (iii) state and local government spending versus federal government spending. The level multipliers from each specification, despite applying a shock to the same total government spending, differ across decompositions. This indicates that the aggregate spending multiplier is not independent of the underlying mix of government purchases.

Turning to the reallocation shocks, a shift toward government investment from government consumption produces a multiplier of approximately 2 on impact. Although the output effect is concentrated in the short run, it underscores the greater productivity of public investment relative to public consumption. A reallocation toward defense spending does not produce a significant change in output, suggesting that the defense versus non-defense distinction is less consequential

for GDP. A reallocation toward state and local government spending raises output significantly, with a five-year multiplier of 2.06, consistent with state and local governments directing a larger proportion of spending toward infrastructure and construction, where demand effects tend to be stronger. These findings suggest that a sustained shift in the allocation of spending toward more productive components can generate meaningful output gains, even without an increase in total government spending.

Our results suggest that a deeper understanding of how government purchases are allocated across categories can enhance fiscal policy effectiveness. While the framework can accommodate additional components, the number of shares that can be included is limited in practice by parameter proliferation and collinearity among the shares. Future research should consider incorporating multiple components of government spending simultaneously, as a component that appears productive at one level of disaggregation may contain sub-components with offsetting effects. Additionally, examining how shifts in the composition of government expenditures over time have influenced the effectiveness of fiscal policy could provide further insights for policy design.

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Figures and Tables

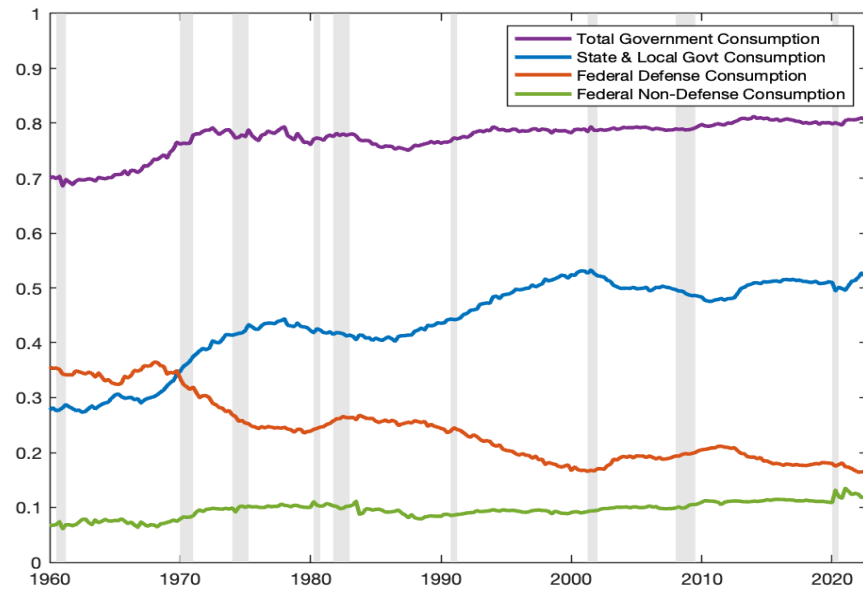


Figure 1.1: Components of Government Consumption Fraction of Total Government Spending

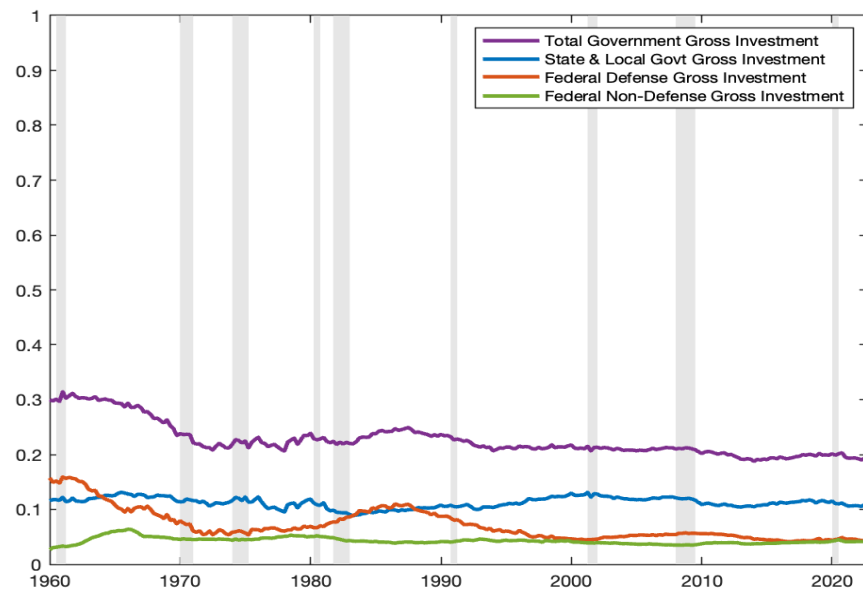


Figure 1.2: Components of Government Gross Investment Fraction of Total Government Spending

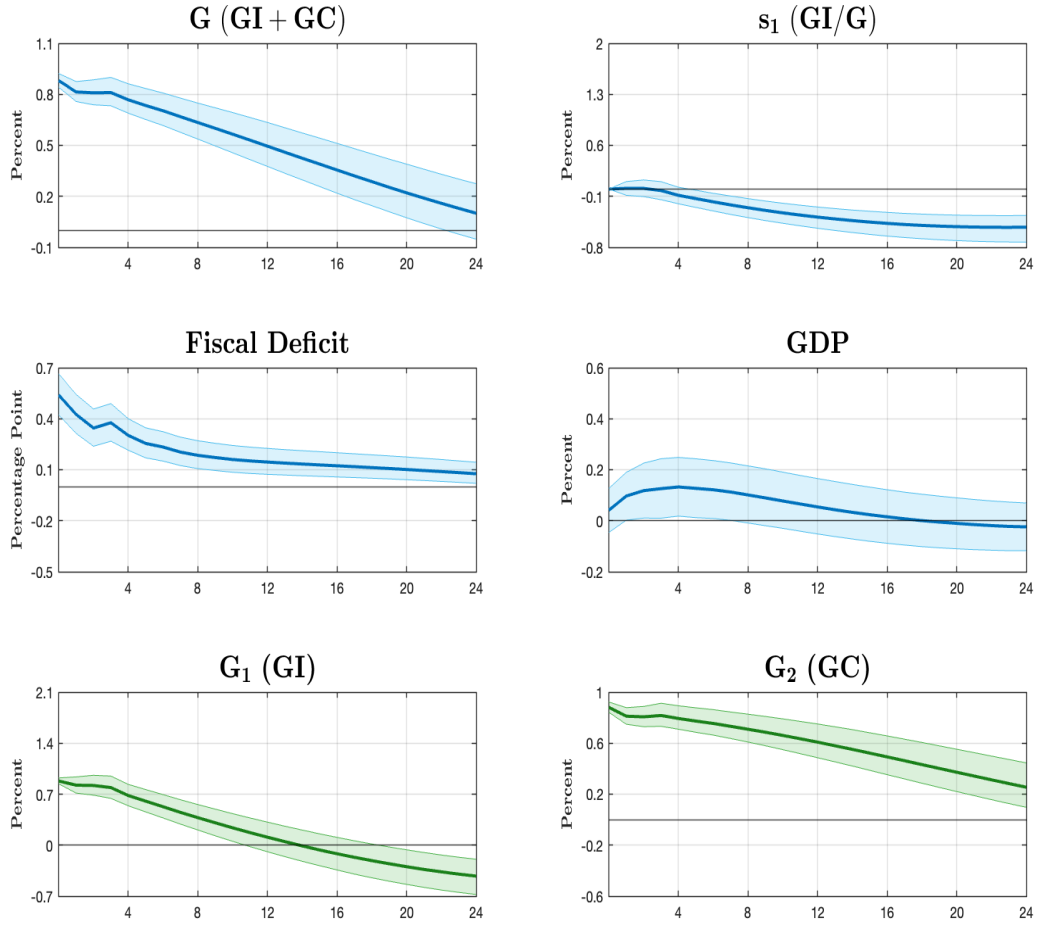


Figure 2.1: Impulse responses to a level shock (shock to G). Here, the level (G) represents total government spending, which consists of government investment (GI) and government consumption (GC), while the share (s_1) is defined as the ratio of government investment (GI) to total government spending (G). Blue and green solid lines represent medians, and the shaded area represents the 68% credible bands.

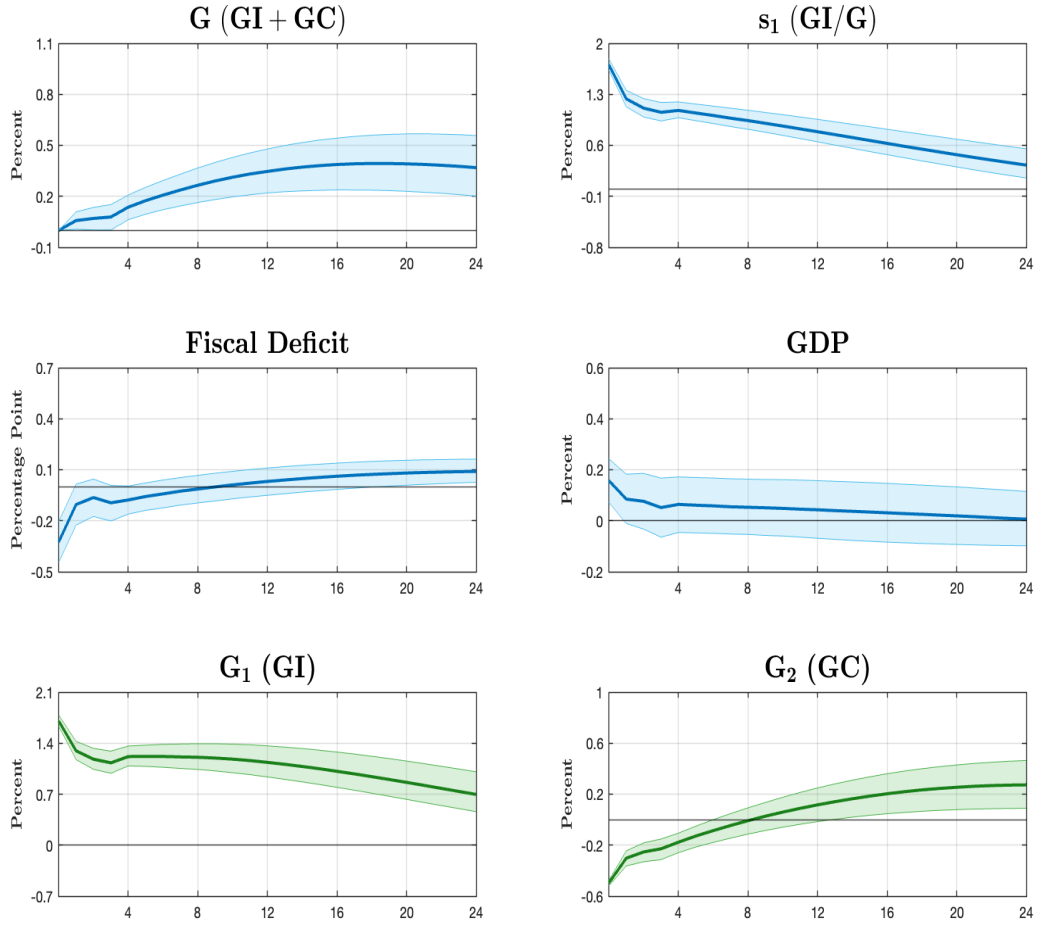


Figure 2.2: Impulse responses to a reallocation shock (shock to s_1). Here, the level (G) represents total government spending, which consists of government investment (GI) and government consumption (GC), while the share (s_1) is defined as the ratio of government investment (GI) to total government spending (G). Blue and green solid lines represent medians, and the shaded area represents the 68% credible bands.

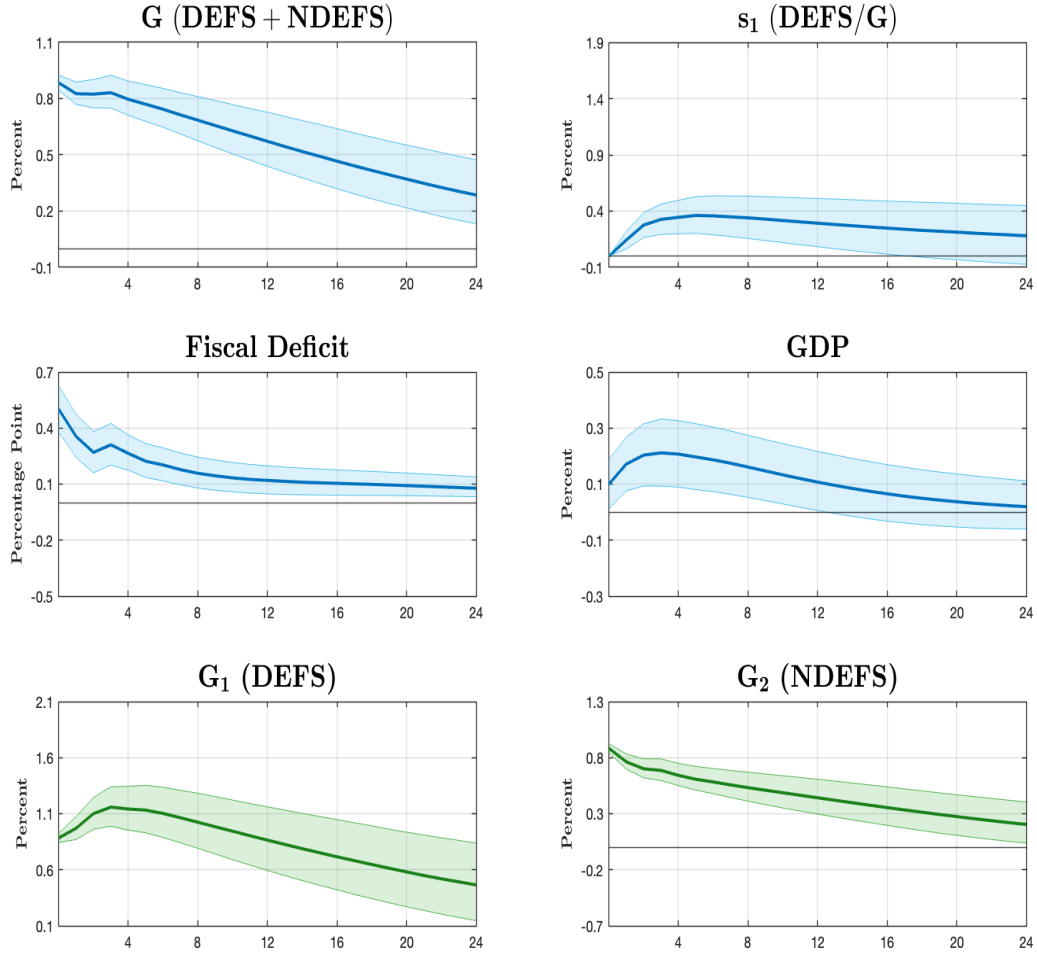


Figure 3.1: Impulse responses to a level shock (shock to G). Here, the level (G) represents total government spending, which consists of defense spending ($DEFS$) and non-defense spending ($NDEFS$), while the share (s_1) is defined as the ratio of defense spending (GI) to total government spending (G). Blue and green solid lines represent medians, and the shaded area represents the 68% credible bands.

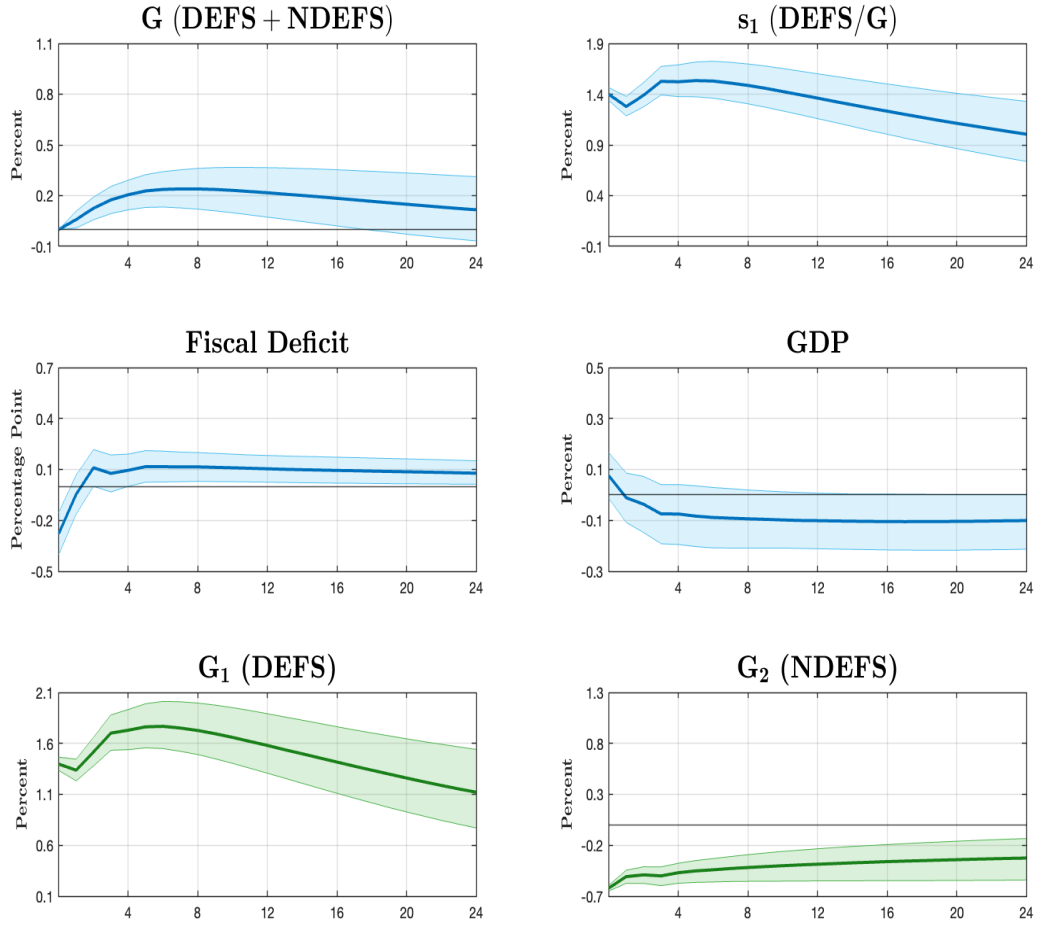


Figure 3.2: Impulse responses to a reallocation shock (shock to s_1). Here, the level (G) represents total government spending, which consists of defense spending ($DEFS$) and non-defense spending ($NDEFS$), while the share (s_1) is defined as the ratio of defense spending (GI) to total government spending (G). Blue and green solid lines represent medians, and the shaded area represents the 68% credible bands.

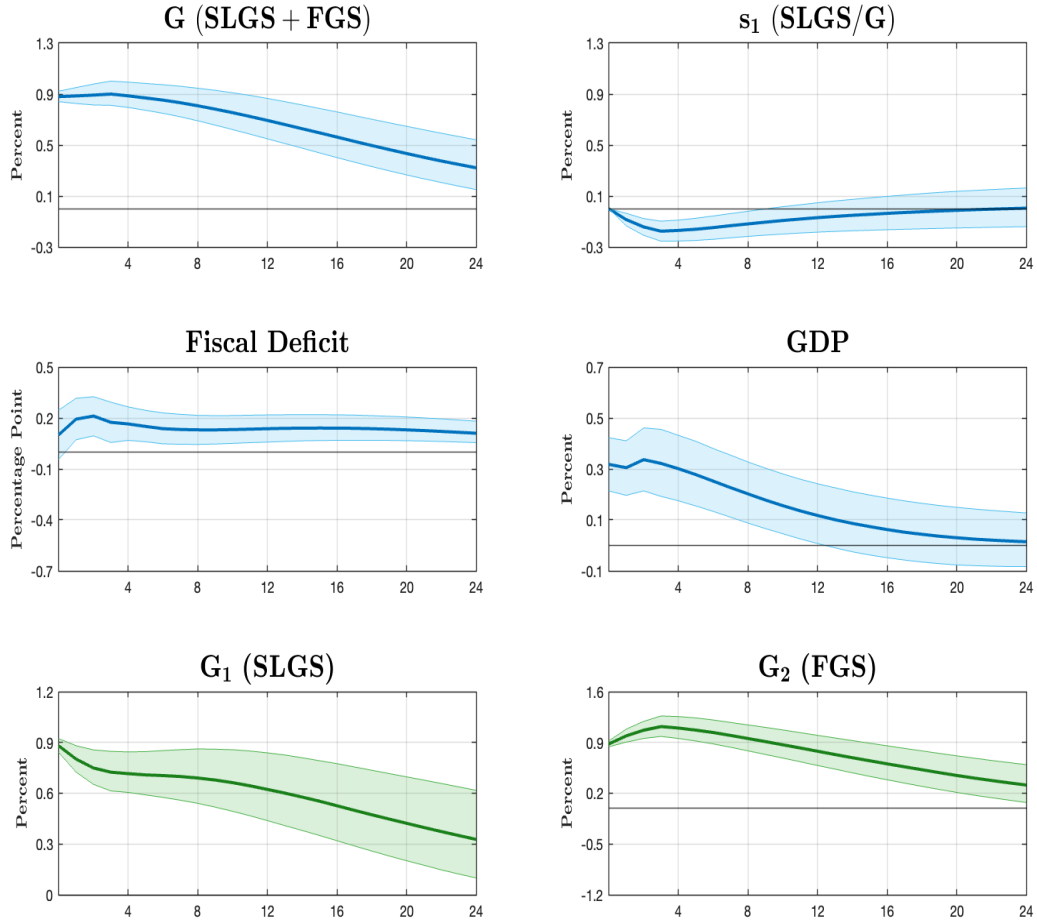


Figure 4.1: Impulse responses to a level shock (shock to G). Here, the level (G) represents total government spending, which consists of state and local government spending ($SLGS$) and federal government spending (FGS), while the share (s_1) is defined as the ratio of state-local spending ($SLGS$) to total government spending (G). Blue and green solid lines represent medians, and the shaded area represents the 68% credible bands.

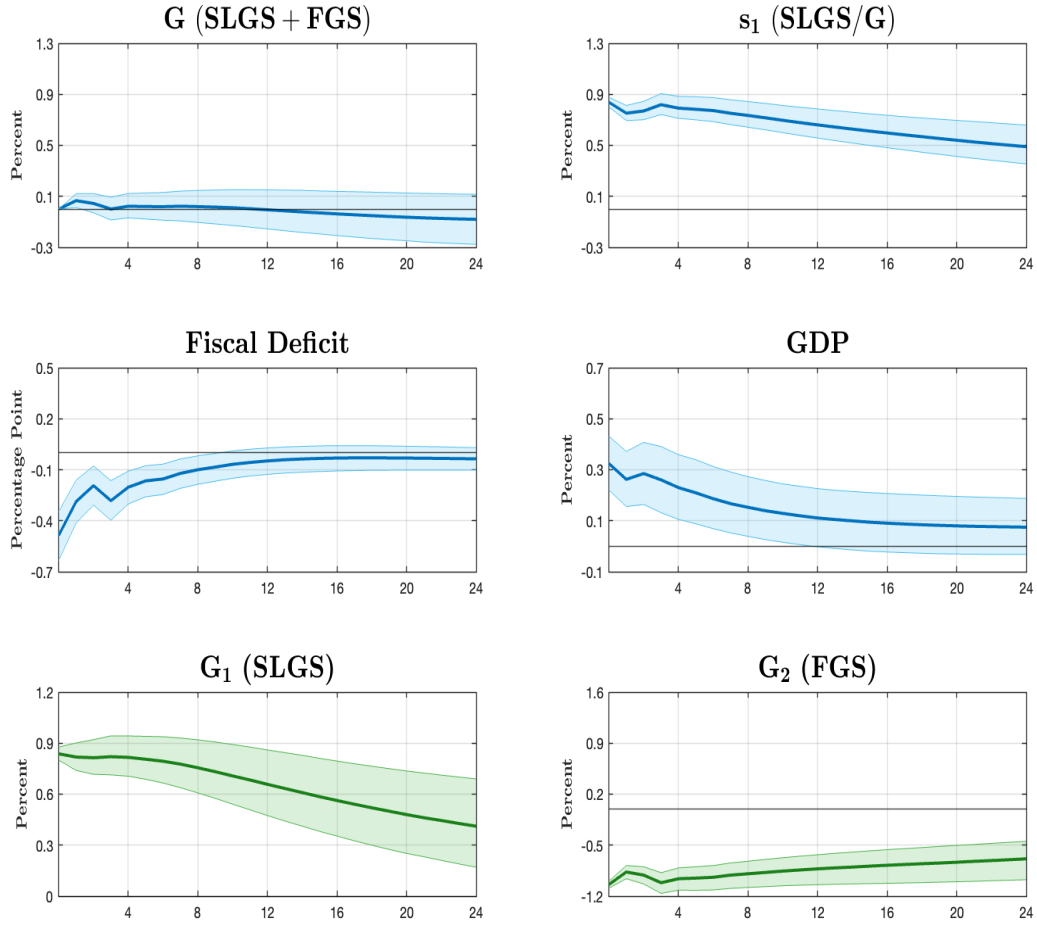


Figure 4.2: Impulse responses to a reallocation shock (shock to s_1). Here, the level (G) represents total government spending, which consists of state and local government spending ($SLGS$) and federal government spending (FGS), while the share (s_1) is defined as the ratio of state-local spending ($SLGS$) to total government spending (G). Blue and green solid lines represent medians, and the shaded area represents the 68% credible bands.

Table 1: Multipliers - Government Investment and Consumption

This table presents the cumulative multipliers for GDP across the full sample at different horizons with 68% credible bands in brackets. Here, G represents the sum of government investment (GI) and government consumption (GC), with s_1 being the ratio of government investment (GI) to total government spending (G). Columns (1) and (2) represent multipliers in response to a level shock (shock to G) and a reallocation shock (shock to s_1), respectively, generated from the VAR specification (2).

Horizon	Level Shock (1)	Reallocation Shock (2)
On Impact	0.23 [-0.26, 0.72]	2.08 [0.93, 3.21]
1 quarter	0.41 [-0.09, 0.89]	1.81 [0.56, 3.08]
4 quarters	0.63 [0.05, 1.18]	1.50 [-0.13, 3.10]
8 quarters	0.71 [0.08, 1.31]	1.31 [-0.42, 2.96]
12 quarters	0.70 [0.02, 1.31]	1.20 [-0.56, 2.86]
16 quarters	0.64 [-0.09, 1.28]	1.10 [-0.72, 2.78]
20 quarters	0.58 [-0.25, 1.26]	1.01 [-0.88, 2.76]
24 quarters	0.52 [-0.42, 1.24]	0.94 [-1.08, 2.73]

Table 2: Multipliers - Defense and Non-Defense Spending

This table presents the cumulative multipliers for GDP across the full sample at different horizons with 68% credible bands in brackets. Here, G represents the sum of defense spending ($DEFS$) and non-defense spending ($NDEFS$), with s_1 being the ratio of defense spending ($DEFS$) to total government spending (G). Columns (1) and (2) represent multipliers in response to a level shock (shock to G) and a reallocation shock (shock to s_1), respectively, generated from the VAR specification (2).

Horizon	Level Shock (1)	Reallocation Shock (2)
On Impact	0.56 [0.06, 1.06]	0.93 [-0.19, 2.08]
1 quarter	0.79 [0.29, 1.28]	0.41 [-0.70, 1.53]
4 quarters	1.07 [0.50, 1.62]	-0.29 [-1.39, 0.82]
8 quarters	1.14 [0.53, 1.70]	-0.58 [-1.67, 0.49]
12 quarters	1.11 [0.49, 1.65]	-0.73 [-1.81, 0.32]
16 quarters	1.05 [0.42, 1.59]	-0.84 [-1.94, 0.20]
20 quarters	0.99 [0.34, 1.54]	-0.93 [-2.07, 0.12]
24 quarters	0.93 [0.26, 1.50]	-1.01 [-2.20, 0.07]

Table 3: Multipliers - State-Local and Federal Spending

This table presents the cumulative multipliers for GDP across the full sample at different horizons with 68% credible bands in brackets. Here, G represents the sum of state and local government spending ($SLGS$) and federal government spending (FGS), with s_1 being the ratio of state and local government spending ($SLGS$) to total government spending (G). Columns (1) and (2) represent multipliers in response to a level shock (shock to G) and a reallocation shock (shock to s_1), respectively, generated from the VAR specification (2).

Horizon	Level Shock (1)	Reallocation Shock (2)
On Impact	1.79 [1.21, 2.38]	3.51 [2.38, 4.63]
1 quarter	1.75 [1.20, 2.29]	3.21 [2.14, 4.27]
4 quarters	1.77 [1.18, 2.34]	3.01 [1.88, 4.09]
8 quarters	1.62 [1.04, 2.16]	2.60 [1.51, 3.61]
12 quarters	1.45 [0.89, 1.96]	2.33 [1.26, 3.25]
16 quarters	1.30 [0.74, 1.81]	2.16 [1.08, 3.04]
20 quarters	1.18 [0.59, 1.71]	2.06 [0.95, 2.92]
24 quarters	1.10 [0.47, 1.64]	2.01 [0.86, 2.85]

Appendix

A Variable Definitions

Variable Definitions

1. **Government Spending Components:** Quarterly data on government purchases are sourced from Table 3.9.5 of the National Income and Product Accounts (NIPA), provided by the Bureau of Economic Analysis. The underlying components are:

- (a) Federal Defense Consumption Expenditures (DEFC) - Row 18
- (b) Federal Defense Gross Investment (DEFI) - Row 19
- (c) Federal Non-Defense Consumption Expenditures (NDEFC) - Row 26
- (d) Federal Non-Defense Gross Investment (NDEFI) - Row 27
- (e) State and Local Government Consumption Expenditures (SLGC) - Row 34,
- (f) State and Local Government Gross Investment (SLGI) - Row 35

All series are converted into real per capita terms using the GDP deflator and population. Total government spending (G) is defined as the sum of all six components. The expenditure categories used in the analysis are constructed as follows:

- Government consumption: the sum of federal defense consumption, federal non-defense consumption, and state and local government consumption.
- Government investment: the sum of federal defense investment, federal non-defense investment, and state and local government investment.
- Defense spending: the sum of federal defense consumption and federal defense investment.
- Non-defense spending: the sum of federal non-defense consumption and investment, and state and local government consumption and investment.
- Federal government spending: the sum of all federal defense and non-defense components.

- State and local government spending: the sum of state and local government consumption and investment.
2. **Population:** Quarterly population data, including resident population plus armed forces overseas from the Federal Reserve Board of St Louis website (B230RC0Q173SBEA).
 3. **Nominal GDP:** This metric is available from Line 1 of Table 1.1.5 in the National Income and Product Accounts (NIPA). Real per capita values are calculated by dividing nominal values by both the GDP deflator and the population figure.
 4. **Implicit Price Deflator for GDP:** This is sourced from Row 1 of Table 1.1.9 of National Income and Product Accounts (NIPA).
 5. **Fiscal Deficit:** Fiscal deficit is defined as the difference between federal government expenditures and federal government receipts as a percentage of nominal GDP. The data on receipts (W018RC1Q027SBEA) and expenditures (W019RCQ027SBEA) are obtained from the Federal Reserve Board of St Louis website.

B Bayesian Estimation

We take a Bayesian perspective to estimate the above reduced-form VAR, treating the model's parameters $\Phi(L)$ and Q as random variables (Bernanke et al., 2005). Define $Y_t = [G_t \ s_{1,t} \ Z_t']'$ as an $N \times 1$ vector. Rewriting the VAR specification (2) as,

$$Y_t = C + \Phi(L)Y_{t-1} + v_t \quad (1)$$

Further, if we assume p lags and define $X_t = (1, Y_{t-1}', \dots, Y_{t-p}')$, then

$$Y_t = X_t \Phi^* + v_t \quad (2)$$

where Φ^* is a $(1 + N \times p) \times N$ matrix of VAR coefficients and T is the sample size.

Since Q is positive definite, it can be decomposed into unique matrices L and D such that $Q^{-1} = L'D^{-1}L$ (Chan and Jeliazkov, 2009). L is a lower triangular matrix with ones on the diagonal and D is a diagonal matrix with positive diagonal elements. Let the diagonal elements in D be denoted as γ_n , $n = 1, \dots, N$ and let $a_{n,m}$, $1 \leq m < n \leq N$ denote the elements of L .

$$D \equiv \begin{bmatrix} \gamma_1 & 0 & \cdots & 0 \\ 0 & \gamma_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \gamma_N \end{bmatrix}; \quad L \equiv \begin{bmatrix} 1 & 0 & \cdots & 0 \\ a_{21} & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ a_{N1} & a_{N2} & \cdots & 1 \end{bmatrix} \quad (3)$$

This decomposition is central to our identification strategy. We impose $a_{21} = 0$, which ensures that total spending (G) does not contemporaneously affect the share (s_1). This restriction ensures that the two variables are contemporaneously independent.

The priors are given as follows:

$$\begin{aligned} \gamma_n &\overset{Ind}{\sim} IG\left(\frac{1+n}{2}, \frac{q_n}{2}\right), \quad n = 1, \dots, N \\ a_n &\overset{Ind}{\sim} N(0, A_0), \quad n = 3, \dots, N \\ \text{vec}(\Phi^*)|Q &\sim N(\text{vec}(\Phi_0^*), Q \otimes \Omega_0) \end{aligned} \quad (4)$$

The diagonal elements q_n are set to the residual variances of the corresponding p -lag univariate

autoregressions, σ_n^2 (Kadiyala and Karlsson, 1997). The element a_{21} is set to zero with probability one to impose the identifying restriction. For the remaining off-diagonal elements a_n , where $n = 3, \dots, N$, the prior mean is set to zero with a large variance $A_0 = 10^3 \cdot I_N$, ensuring that these elements are unrestricted.

For the VAR coefficients, we employ a Minnesota prior. The prior mean is set to $\Phi_0^* = 0$. The prior variance Ω_0 is a $(1 + pN) \times (1 + pN)$ matrix with diagonal elements set such that the prior variance of the coefficient on the l th lag of the m th variable in the n th equation equals $\sigma_n^2 / l \sigma_m^2$. This specification embeds the belief that coefficients on longer lags are more likely to be zero and that own lags are more informative than lags of other variables. We set the hyperparameters $\lambda_2 = \lambda_3 = \lambda_4 = 1$, following Bernanke et al. (2005), but adopt a tighter overall shrinkage of $\lambda_1 = 0.2$ to regularize the posterior moderately toward the prior.

The posterior is given as:

$$\begin{aligned} \gamma_n | X, Y &\stackrel{Ind}{\sim} IG \left(\frac{1 + n + T}{2}, \frac{\bar{q}_n}{2} \right) \\ a_n | X, Y, \gamma_n &\stackrel{Ind}{\sim} N(\bar{a}_n, \bar{A}_n) \\ vec(\Phi^*) | X, Y, Q &\sim N(vec(\bar{\Phi}^*), Q \otimes \bar{\Omega}) \end{aligned} \quad (5)$$

where $\bar{q}_n$ is the n^{th} diagonal element of $\bar{Q}$.

$$\begin{aligned} \bar{Q} &= Q_0 + \hat{v}'\hat{v} + (\hat{\Phi}^* - \Phi_0^*)' \left[\Omega_0 + (X'X)^{-1} \right]^{-1} (\hat{\Phi}^* - \Phi_0^*) \\ \bar{A}_n &= \gamma_n (A_0^{-1} + \hat{v}_n' \hat{v}_n)^{-1} \\ \bar{a}_n &= \bar{A}_n \left(\gamma_n^{-1} \hat{v}_n' \hat{V}_n \right) \\ \hat{v}_n &= [\hat{V}_1 : \dots : \hat{V}_n] \\ \hat{V}_n &= (\hat{V}_{1n}, \hat{V}_{2n}, \dots, \hat{V}_{Tn})' \end{aligned} \quad (6)$$

Also, we have,

$$\begin{aligned} \bar{\Phi}^* &= \bar{\Omega} \left(\Omega_0^{-1} \Phi_0^* + X'X \hat{\Phi}^* \right) \\ \bar{\Omega} &= (\Omega_0^{-1} + X'X)^{-1} \end{aligned} \quad (7)$$

Here, $\hat{\Phi}^*$ and $\hat{v}$ refer to the OLS estimates of the VAR coefficients and the reduced-form residuals for the VAR model.

Estimation proceeds via a Gibbs sampler that iterates over the following steps. First, condi-

tional on the current draw of Q , draw Φ^* from its conditional posterior. We discard draws of Φ^* with roots outside the unit circle, as the VAR model would be unstable (Bernanke et al., 2005). Second, conditional on the new draw of Φ^* , draw the diagonal elements γ_n and the unrestricted off-diagonal elements a_n of the LDL decomposition from their respective conditional posteriors, with a_{21} held fixed at zero throughout. These draws implicitly define the new Q . For every iteration, we compute impulse responses to the level and reallocation shocks as detailed in Section 3.4, and the corresponding multipliers as outlined in Section 3.5. We run the sampler for 30,000 iterations and discard the first 15,000 as burn-in.

C Discounted Multipliers

Table C.1: Present Value Multipliers - Government Investment and Consumption

This table presents the discounted cumulative multipliers for GDP across the full sample at different horizons with 68% credible bands in brackets. Here, G represents the sum of government investment (GI) and government consumption (GC), with s_1 being the ratio of government investment (GI) to total government spending (G). Columns (1) and (2) represent multipliers in response to a level shock (shock to G) and a reallocation shock (shock to s_1), respectively, generated from the VAR specification (2).

Horizon	Level Shock (1)	Reallocation Shock (2)
On Impact	0.23 [-0.26, 0.72]	2.08 [0.93, 3.21]
1 quarter	0.41 [-0.09, 0.90]	1.81 [0.54, 3.09]
4 quarters	0.64 [0.05, 1.21]	1.47 [-0.18, 3.11]
8 quarters	0.73 [0.08, 1.34]	1.27 [-0.49, 2.95]
12 quarters	0.70 [-0.01, 1.34]	1.14 [-0.68, 2.83]
16 quarters	0.63 [-0.18, 1.31]	1.02 [-0.88, 2.78]
20 quarters	0.53 [-0.44, 1.27]	0.93 [-1.15, 2.77]
24 quarters	0.42 [-0.81, 1.28]	0.80 [-1.46, 2.77]

Table C.2: Present Value Multipliers - Defense and Non-Defense Spending

This table presents the discounted cumulative multipliers for GDP across the full sample at different horizons with 68% credible bands in brackets. Here, G represents the sum of defense spending ($DEFS$) and non-defense spending ($NDEFS$), with s_1 being the ratio of defense spending ($DEFS$) to total government spending (G). Columns (1) and (2) represent multipliers in response to a level shock (shock to G) and a reallocation shock (shock to s_1), respectively, generated from the VAR specification (2).

Horizon	Level Shock (1)	Reallocation Shock (2)
On Impact	0.56 [0.06, 1.06]	0.93 [-0.19, 2.08]
1 quarter	0.80 [0.30, 1.29]	0.40 [-0.72, 1.52]
4 quarters	1.08 [0.51, 1.64]	-0.32 [-1.43, 0.79]
8 quarters	1.15 [0.53, 1.72]	-0.63 [-1.73, 0.45]
12 quarters	1.11 [0.48, 1.67]	-0.79 [-1.89, 0.27]
16 quarters	1.03 [0.37, 1.59]	-0.93 [-2.06, 0.14]
20 quarters	0.94 [0.23, 1.53]	-1.04 [-2.26, 0.06]
24 quarters	0.85 [0.07, 1.48]	-1.15 [-2.46, 0.01]

Table C.3: Present Value Multipliers - State-Local and Federal Spending

This table presents the discounted cumulative multipliers for GDP across the full sample at different horizons with 68% credible bands in brackets. Here, G represents the sum of state and local government spending ($SLGS$) and federal government spending (FGS), with s_1 being the ratio of state and local government spending ($SLGS$) to total government spending (G). Columns (1) and (2) represent multipliers in response to a level shock (shock to G) and a reallocation shock (shock to s_1), respectively, generated from the VAR specification (2).

Horizon	Level Shock (1)	Reallocation Shock (2)
On Impact	1.79	3.51
	[1.21, 2.38]	[2.38, 4.63]
1 quarter	1.75	3.20
	[1.20, 2.29]	[2.14, 4.26]
4 quarters	1.77	2.99
	[1.18, 2.34]	[1.85, 4.08]
8 quarters	1.60	2.54
	[1.02, 2.14]	[1.43, 3.55]
12 quarters	1.39	2.22
	[0.82, 1.91]	[1.12, 3.15]
16 quarters	1.20	2.02
	[0.61, 1.74]	[0.88, 2.92]
20 quarters	1.05	1.91
	[0.39, 1.61]	[0.67, 2.80]
24 quarters	0.92	1.85
	[0.17, 1.54]	[0.50, 2.76]